

Step 4: Two-Way ANOVA (House Type * Area)

Danish Housing Prices · Mathematical Statistics

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```
library(tidyverse)
dk_clean <- readRDS("../..data/dk_clean.rds")

# Restrict to 4 areas for a balanced comparison
dk_anova <- dk_clean %>%
  filter(area %in% c("Capital, Copenhagen", "North Zealand",
                    "East & mid jutland", "South jutland")) %>%
  droplevels()

cat("Observations per cell (house_type × area):\n")
print(table(dk_anova$house_type, dk_anova$area))
```

Observations per cell (house_type × area):

	Capital, Copenhagen	East & mid jutland	North Zealand
Apartment	6554	2602	818
Summerhouse	128	970	1360
Townhouse	2205	1554	974
Villa	5276	8609	3296

	South jutland
Apartment	2502
Summerhouse	2174
Townhouse	1342
Villa	12333

1 Statistical Model — Five-Model Hierarchy

With row factor = house type ($r = 4$ levels) and column factor = area ($s = 4$ levels):

M1 (interaction): $X_{ijk} \sim \mathcal{N}(\mu_{ij}, \sigma^2)$

M2 (additive): $X_{ijk} \sim \mathcal{N}(\alpha_i + \beta_j, \sigma^2)$

M3r (row only): $X_{ijk} \sim \mathcal{N}(\alpha_i, \sigma^2)$

M3c (column only): $X_{ijk} \sim \mathcal{N}(\beta_j, \sigma^2)$

M_∞ (common mean): $X_{ijk} \sim \mathcal{N}(\mu, \sigma^2)$

2 Variance Homogeneity — Bartlett's Test on Cells

For two-way ANOVA the relevant groups are the **crossed cells** (type $\times$ area). The course requires `interaction()` to define them correctly — the colon operator alone does not work for `bartlett.test`.

```
cat("Per-cell standard deviations:\n")
cell_sds <- tapply(dk_anova$log_price,
                  interaction(dk_anova$house_type, dk_anova$area), sd)
print(round(sort(cell_sds), 3))

cat("\nMax / Min SD ratio:", round(max(cell_sds) / min(cell_sds), 2), "\n\n")

cat("--- Bartlett's test on cells ---\n")
bartlett.test(log_price ~ interaction(house_type, area), data = dk_anova)

cat("\nSame large-n caveat as Step 3: check the ratio above.\n")
```

Per-cell standard deviations:

Townhouse.North Zealand	Townhouse.Capital, Copenhagen
0.498	0.576
Apartment.North Zealand	Summerhouse.South jutland
0.585	0.620
Villa.South jutland	Townhouse.East & mid jutland
0.642	0.643
Summerhouse.East & mid jutland	Summerhouse.Capital, Copenhagen
0.644	0.644
Villa.North Zealand	Apartment.Capital, Copenhagen
0.649	0.668

Townhouse.South jutland	Villa.Capital, Copenhagen
0.677	0.683
Summerhouse.North Zealand	Villa.East & mid jutland
0.716	0.721
Apartment.East & mid jutland	Apartment.South jutland
0.739	0.746

Max / Min SD ratio: 1.5

--- Bartlett's test on cells ---

Bartlett test of homogeneity of variances

data: log_price by interaction(house_type, area)
 Bartlett's K-squared = 573.96, df = 15, p-value < 2.2e-16

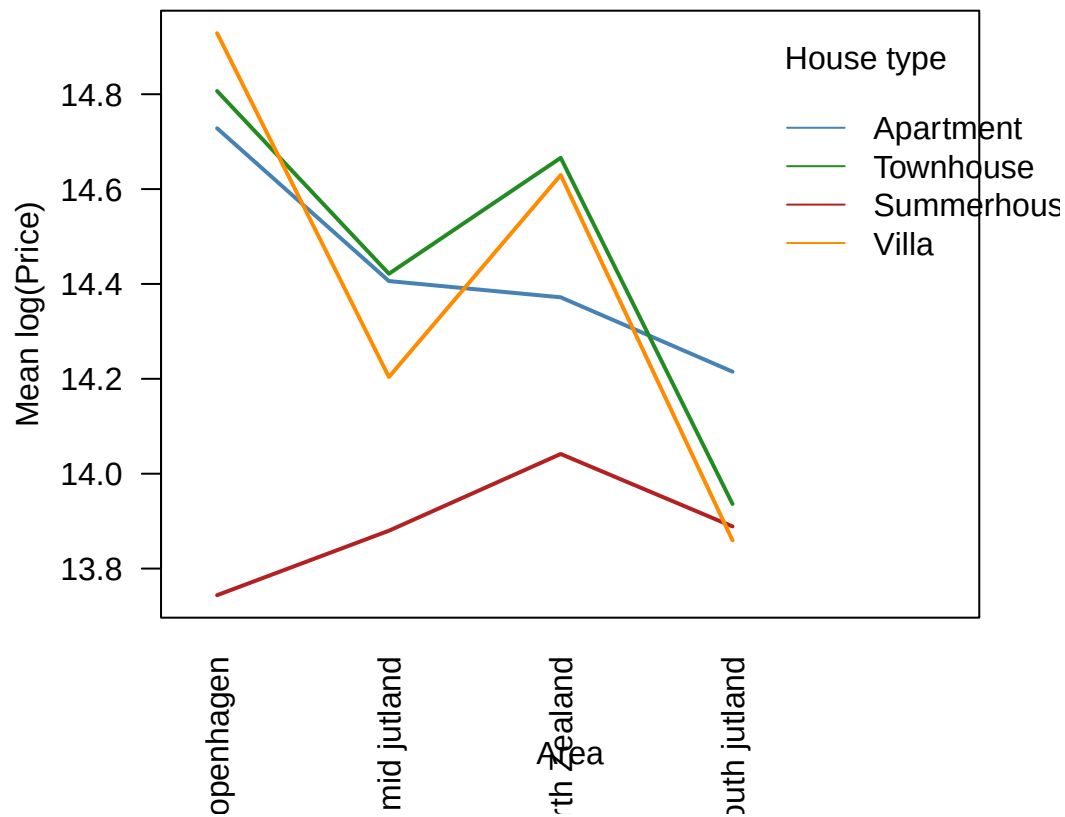
Same large-n caveat as Step 3: check the ratio above.

3 Interaction Plot

Parallel lines indicate no interaction (additive model adequate). Non-parallel lines suggest the area effect differs by house type.

```
interaction.plot(
  x.factor      = dk_anova$area,
  trace.factor  = dk_anova$house_type,
  response      = dk_anova$log_price,
  col           = c("steelblue", "firebrick", "forestgreen", "darkorange"),
  lty           = 1, lwd = 2,
  main          = "Interaction Plot: Area × House Type",
  xlab          = "Area",
  ylab          = "Mean log(Price)",
  trace.label    = "House type",
  las           = 2
)
```

Interaction Plot: Area × House Type



4 Model Fitting and Tests

4.1 Test 1 — Can we remove the interaction? (M1 → M2)

$$H_0 : \mu_{ij} = \alpha_i + \beta_j \text{ for all } i, j \quad \text{vs} \quad H_1 : \text{interaction present}$$

```
M1_2way <- lm(log_price ~ house_type * area, data = dk_anova)
M2_2way <- lm(log_price ~ house_type + area, data = dk_anova)

cat("--- F-test: M2 vs M1 (remove interaction) ---\n")
anova(M2_2way, M1_2way)
```

```

--- F-test: M2 vs M1 (remove interaction) ---
Analysis of Variance Table

Model 1: log_price ~ house_type + area
Model 2: log_price ~ house_type * area
   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1  52690 24425
2  52681 23636   9    789.23 195.46 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

4.2 Test 2 — House type effect? (M3c → M2)

$H_0 : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$ (no house type effect given additive model)

```

M3c_2 <- lm(log_price ~ area, data = dk_anova)

cat("--- F-test: M3c vs M2 (house_type effect) ---\n")
anova(M3c_2, M2_2way)

```

```

--- F-test: M3c vs M2 (house_type effect) ---
Analysis of Variance Table

Model 1: log_price ~ area
Model 2: log_price ~ house_type + area
   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1  52693 24803
2  52690 24425   3    377.89 271.73 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

4.3 Test 3 — Area effect? (M3r → M2)

$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4$ (no area effect given additive model)

```

M3r_2 <- lm(log_price ~ house_type, data = dk_anova)

cat("--- F-test: M3r vs M2 (area effect) ---\n")
anova(M3r_2, M2_2way)

```

```

--- F-test: M3r vs M2 (area effect) ---
Analysis of Variance Table

Model 1: log_price ~ house_type
Model 2: log_price ~ house_type + area
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1  52693 29699
2  52690 24425  3    5273.7 3792.2 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

5 Additive Model — Summary

```

summary(M2_2way)

cat("\nResidual sigma s =", round(summary(M2_2way)$sigma, 4), "\n")

```

```

Call:
lm(formula = log_price ~ house_type + area, data = dk_anova)

Residuals:
    Min       1Q   Median       3Q      Max
-2.3959 -0.4510  0.0277  0.4668  2.5560

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    14.828133   0.006988 2122.064 < 2e-16 ***
house_typeSummerhouse -0.324654   0.012370  -26.245 < 2e-16 ***
house_typeTownhouse    0.024388   0.010736   2.272  0.0231 *
house_typeVilla       -0.061175   0.007698  -7.947 1.95e-15 ***
areaEast & mid jutland -0.525728   0.008454 -62.188 < 2e-16 ***
areaNorth Zealand    -0.253711   0.010699 -23.714 < 2e-16 ***
areaSouth jutland     -0.833295   0.008138 -102.395 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6809 on 52690 degrees of freedom
Multiple R-squared:  0.2213,    Adjusted R-squared:  0.2212
F-statistic: 2496 on 6 and 52690 DF,  p-value: < 2.2e-16

```

Residual sigma s = 0.6809

6 Cell Mean Estimates (Additive Model)

```
# Reconstruct fitted means for each cell
grid <- expand.grid(
  house_type = levels(dk_anova$house_type),
  area       = levels(dk_anova$area)
)
grid$fitted_log <- predict(M2_2way, newdata = grid)
grid$fitted_dkk <- round(exp(grid$fitted_log))

cat("Fitted median prices (DKK) under additive model:\n")
mat <- matrix(format(grid$fitted_dkk, big.mark = ","),
              nrow = 4, ncol = 4,
              dimnames = list(levels(dk_anova$house_type),
                              levels(dk_anova$area)))
print(mat)
```

Fitted median prices (DKK) under additive model:

	Capital, Copenhagen	East & mid jutland	North Zealand	South jutland
Apartment	"2,752,811"	"1,627,255"	"2,135,949"	"1,196,413"
Summerhouse	"1,989,669"	"1,176,143"	"1,543,816"	" 864,740"
Townhouse	"2,820,771"	"1,667,427"	"2,188,681"	"1,225,949"
Villa	"2,589,456"	"1,530,691"	"2,009,199"	"1,125,416"

7 Interpretation

The additive model is appropriate if the interaction test is not significant, meaning the **price premium of one house type over another is the same across all areas**. If interaction is significant, each type $\times$ area combination needs its own mean, and the simpler additive story breaks down.